JetBrains Developer Ecosystem Survey 2024

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Insights from 23,262 developers worldwide on programming language usage, satisfaction, adoption intent, and domain specialization · 2017–2024

SURVEY SIZE	YEARS TRACKED	LANGUAGES
23,262 Developers	**8 Years · 2017–2024**	**20 Languages**

KEY FINDING

JavaScript peaked at 70% in 2020 and is declining; Python has nearly closed the gap, rising from 32% to 57% — reshaping the top-5 landscape

Python Closes the Gap: 8 Years of Language Usage Shifts

LANGUAGE	2017	2020	2022	2024	TREND BAR (2024)	DIRECTION
JavaScript Web / Full-stack	65%	**70%**	65%	**61%**		▼ Declining
Python AI / Data / Scripting	32%	55%	53%	**57%**		▼ ↑ Growing
HTML / CSS Web Markup	60%	61%	54%	**51%**		▼ Declining
SQL Data / Databases	42%	56%	49%	**48%**		Plateaued
Java Enterprise / Android	47%	54%	48%	**46%**		▼ Slow decline
TypeScript Web / Full-stack	12%	28%	34%	**37%**		▲ ↑ Tripled
Go Cloud / Infrastructure	8%	19%	19%	**18%**		Plateaued
Rust Systems / Performance	—	7%	9%	**11%**		▲ ↑ Growing
PHP Web Backend	30%	27%	20%	**17%**		▼ ↓ −43%
Kotlin Android / JVM	2%	17%	16%	**14%**		Plateaued

Source: JetBrains Developer Ecosystem Survey 2024 · N=23,262 · Bars scaled to max=80%

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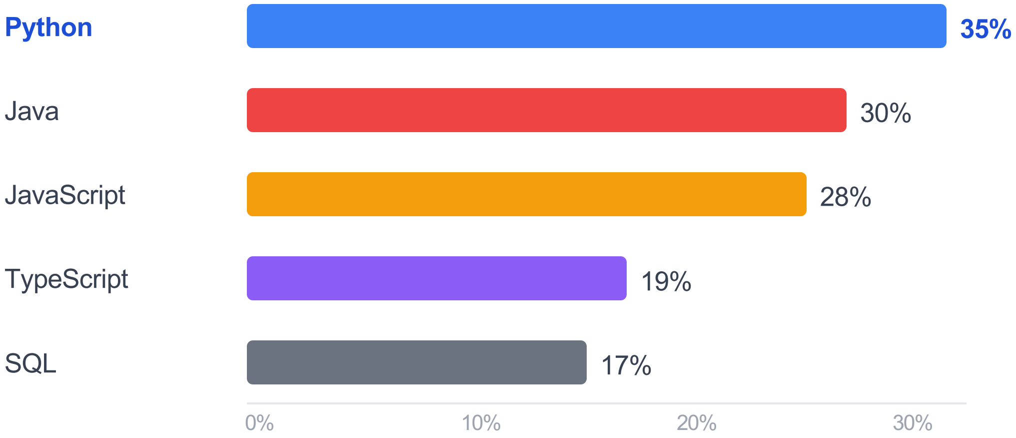
KEY FINDING

Developer ecosystem lock-in is strong — adoption intent stays below 6% while Python emerges as the #1 primary language at 35%

Python Is Now the #1 Primary Language at 35%

Primary Language % of Developers



Strategic implication:
Python's rise to #1 primary language is driven by the AI/ML boom — 31% of AI developers use it

WHAT THIS MEANS FOR HIRING

Python talent is now the most in-demand primary skill. Yet switching costs are high — retrain existing engineers rather than expecting new-hire migration.

Primary Usage vs. Adoption Intent

Developers rarely plan to switch their primary language

LANGUAGE	PRIMARY %	ADOPTION INTENT
Python	35%	6%
Java	30%	4%
JavaScript	28%	~2%
TypeScript	19%	5%
SQL	17%	4%

THE LOCK-IN EFFECT

Adoption intent is below 6% across all languages. Developers don't switch primary stacks easily — ecosystem investment, tooling, and expertise create strong inertia.

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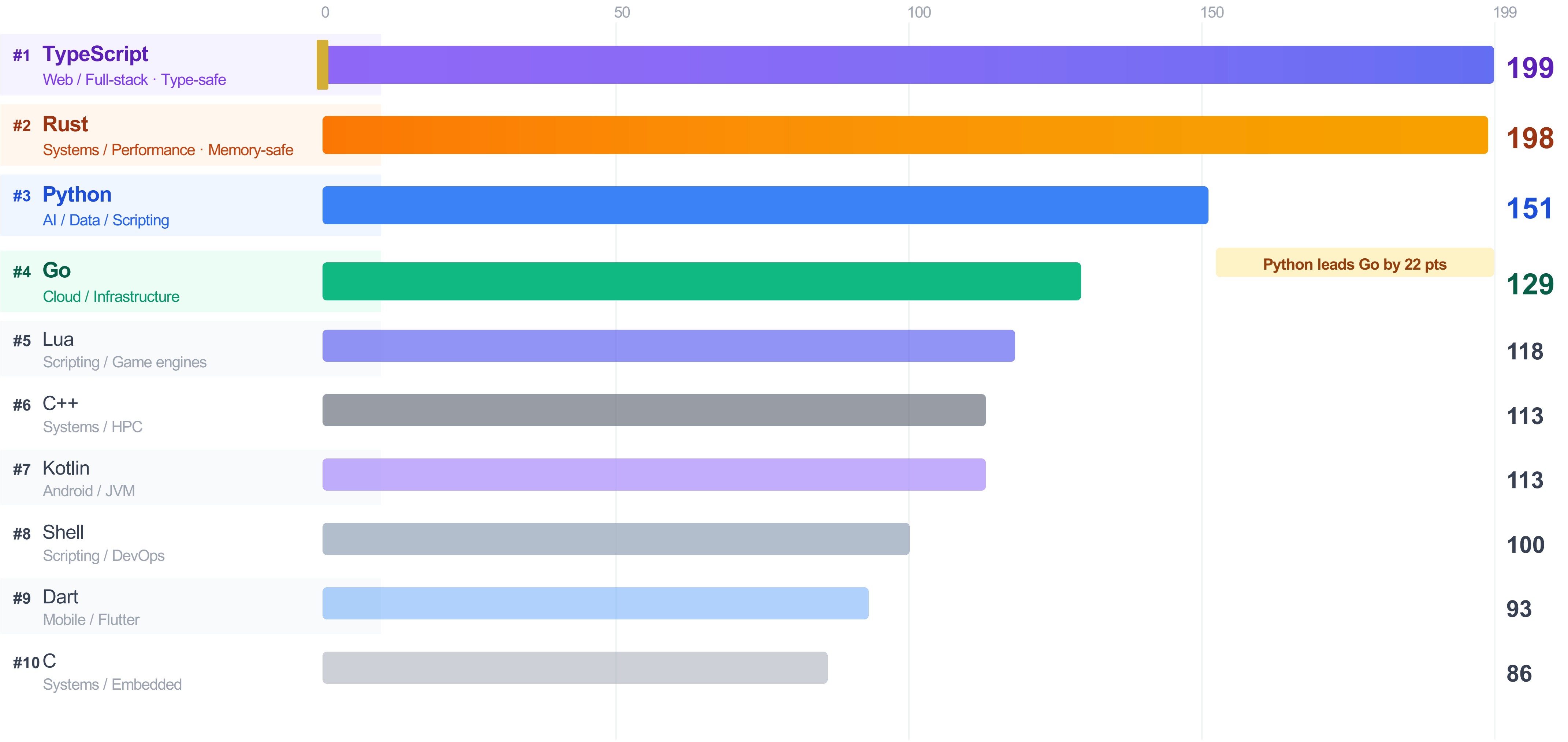
KEY FINDING

Type-safe, modern-tooling languages dominate satisfaction — TypeScript (199) and Rust (198) lead by 32% over Python (151)

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



PATTERN

Top 2 are type-safe languages

TS+Rust gap over Python:

+48

points

Use for retention strategy

KEY FINDING

Language choice is domain-driven, not preference-driven — Python owns AI/data, TypeScript owns UI at 62%, Java owns enterprise services

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

APPLICATION DOMAIN	DOMINANT LANGUAGE	SHARE	INTENSITY	STRATEGIC IMPLICATION
AI / Machine Learning Model training, inference, automation	**Python** TensorFlow · PyTorch · scikit-learn	31%		Hire Python-first for all AI teams
Data Science & Analytics Statistical analysis, visualization, BI	**Python** pandas · NumPy · Jupyter	28%		Python+SQL is the data stack standard
Integrating with APIs & Services REST clients, glue code, automation	**Python** requests · httpx · aiohttp	38%		Python is the universal integration glue
User Interface Development Web frontend, mobile UI, design systems	**TypeScript** React · Vue · Angular · Next.js	62%		**Standardize TypeScript in web teams now**
APIs & Services REST/gRPC APIs, microservices	**Java** Spring Boot · Quarkus · Micronaut	49%		Java remains king for enterprise services
Application Logic & Workflows Business logic, orchestration, backend	**Java** Spring · Jakarta EE · Hibernate	55%		Strong Java moat in enterprise workflows
System Tools & Components OS-level tools, compilers, drivers	**C++** LLVM · Qt · Boost	27%		C++ still dominates; Rust is the challenger

BOTTOM LINE: Python owns 3 of 7 domains (AI, data science, API integration). TypeScript dominates UI at 62%. Java owns enterprise at 49–55%.
Hire and invest by domain — language follows workload, not personal preference.

Source: JetBrains Developer Ecosystem Survey 2024 · Domain percentages represent top-language share of developers in that domain

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STRATEGIC RECOMMENDATION

Five decisions for 2025: invest in Python, standardize TypeScript, bet on Rust/Go for infra, wind down PHP/Ruby, use language satisfaction as a retention lever

What This Means for Your Technology Strategy in 2025

DECISION 01

Invest in Python for AI/ML Teams

Python leads AI/ML (31%), data science (28%), and API integration (38%). It is now the #1 primary language at 35% of developers — driven by the AI boom.

Non-negotiable for AI orgs

DECISION 02

TypeScript: Future of Frontend & Full-Stack

Usage tripled from 12% (2017) to 37% (2024). Dominates UI development at 62%. Tied #1 in satisfaction at score 199 — developers love it and adopt it rapidly.

Standardize TypeScript in web teams

DECISION 03

Rust & Go: Infrastructure Growth Plays

Most-planned-to-adopt languages. Go: 8%→18% (2017–2024), stabilized for cloud-native. Rust: effectively 0%→11%, still climbing for systems, infra, and performance-critical services.

Hire Rust/Go for infra & platform teams

DECISION 04

PHP & Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17% over 8 years (–43%). Ruby fell from 10% to 4%. New developer talent is choosing Python and TypeScript instead.

Factor into long-term talent planning

DECISION 05 — RETENTION LEVER

Satisfaction = Retention: TypeScript (199) & Rust (198) developers are the most engaged

Language choice directly impacts developer happiness. Giving engineers access to modern, type-safe languages reduces attrition risk. Factor this into stack decisions.